

# Micro Eazy is already running. We are pointed at the wrong copy of it.

A probe of Micromart's production ServiceSuite over the Tailscale route, checked line by line against the Micro Eazy Ecosystem Blueprint. The headline: the Fintech entity is live, funded and disbursing — and the LMS bridge is configured for a different deployment that does not contain any of it. **Revised 12 Aug:** four decisions are now locked, two blockers cleared, and the fix is scoped for every lender on the platform rather than Micromart alone.

Probed 12 Aug 2026    Host 100.72.35.56,4230 (Tailscale)    Server services · SQL 2022 16.0.1190.2  
Catalog Serviceconnect + Transactions    Blueprint v1.0, 5 Aug 2026

|                                                                        |                                                                                        |                                                                          |                                                                   |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------|
| <div>TAILSCALE ROUTE</div> <div>Live</div> <div>read+exec proven</div> | <div>FINTECH ENTITY</div> <div>3005</div> <div>MICROMART FINTECH</div>                 | <div>CUSTOMERS PORTED</div> <div>17,016</div> <div>2 Aug, 22:43:07</div> | <div>MICRO EAZY LOANS</div> <div>37</div> <div>33 disbursed</div> |
| <div>DISBURSED TO DATE</div> <div>382k</div> <div>KES, all sent</div>  | <div>YOUR ACCOUNT</div> <div>1 gate</div> <div>unit+limit fixed;<br/>wallet left</div> | <div>VIA LMS BRIDGE</div> <div>0</div> <div>of 37 loans</div>            | <div>SMS + EMAIL</div> <div>Dead</div> <div>since May 2023</div>  |

## 00 Decisions locked, and what they change

Answered 12 Aug. Each one removes a branch from the plan — and one of them widens it.

VERIFIED LIVE

### Demo limit and org unit are fixed

I re-probed rather than take it on trust. Both changes landed at 11 Aug 23:31:19 — and the limit went to 30,000, giving headroom above the 25,000 demo rather than sitting exactly on it.

```
Borrowers.ID 168346
  EntityUnit  129      (was 1)      <- now inside the products' "0,129" scope
  LoanLimit   30,000.00 (was 10,000)

dbo.LoanValidation(168346, 30220, 25000) -> 2      (was 3 "exceeds loan limit")
dbo.LoanValidation(168346, 30220, 5000)  -> 2
dbo.LoanValidation(168346, 30219, 5000)  -> 2
AccountSavings row for 168346           -> none
```

The limit blocker is gone. **All three now fail on code 2 — the wallet — which is the one remaining gate,** and the one decision 4 addresses.

### **SIMPLIFICATION** The old Techcrast box was test; this server is live

That collapses the cross-server posting split entirely. `POSTING_TARGETS` and `getPostingOrg()` in `connections.ts` exist only to read a lender's book from one server and post loans to another. With the test deployment retired, **read and post are the same connection** — so that indirection should be deleted, not repointed. Fewer moving parts, and no way to book into the wrong ledger by misconfiguration.

It also means the `techcrast` org entry — `EntityId 7` on `MICROMART_FINTECH` — currently points at a test box for a lender you describe as live. Techcrast needs a real home before it can transact.

### **SCOPE CHANGE** This is a platform fix, not a Micromart fix

Every defect in sections 05 and 06 is structural, so fixing it per-lender would mean fixing it four or five times. The roster as the code currently defines it:

```
BRIDGED, own server      micromart 3002 · micromart-fintech 3005 · axe 3003 · techcrast 7
BRIDGED, shared server  buysimu 8 · njb 3 · atico 23
NATIVE   (our own book)  hub · Mular Credit Ltd – prospect, not yet an org row
```

Note the hazard that scope exposes: `axe` is registered as `EntityId 3003`, and `3003` on *this* server is Micromart Check off. Entity IDs are only meaningful alongside their connection, yet the registry stores them as bare integers per slug. That is the same class of mistake that pointed Micromart's bridge at a dead deployment.

### **APPROVED** Our app collects the charges by STK before the offer is signed

Confirmed as the design. Section 09 now carries the exact sequence, because the wallet mechanics contain a trap that would have quietly eaten the money.

## 01 The route works, and it is the same route the portal uses

First thing to settle, because everything else depends on it.

The connection string you gave me opens against `services\Serviceconnect` on SQL Server 2022 over the Tailscale address. I ran reads, scalar functions and stored-procedure definition dumps across two databases ( `Serviceconnect` and `Transactions` ) with no tunnelling tricks — the route is not just reachable, it carries real query load.

More useful: `ServiceSuite/appsettings.json` in the portal repo now has that *exact same* string as its single active connection, with the older `102.210.148.124,4230` line commented out directly above it. The portal and I are reading the same database through the same pipe. That half of "connect the suite over Tailscale" is done and proven.

## ONE CAVEAT WORTH NAMING NOW

The portal's `HostSettings.EntityId` is still `3003` (Micromart Check off) with `HostPath: "/axe"`. That is a leftover dev binding, not the Fintech entity. Whoever runs the portal against this DB is looking at the wrong book until that becomes `3005`.

## 02 What is actually in the Fintech entity

Blueprint §2.2 described this as a configuration task not yet started. It has been done — and then some.

`BsEntity` carries four Micromart entities. The one you asked about is **3005 — MICROMART FINTECH**, and it is no longer a shell:

| BSENTITY · MICROMART ESTATE ON THIS SERVER |                          |           |         |                       |
|--------------------------------------------|--------------------------|-----------|---------|-----------------------|
| ID                                         | ENTITY                   | BORROWERS | ACTIVE  | NOTE                  |
| 3002                                       | Micromart Africa Limited | 140,312   | 140,004 | the historic book     |
| 3005                                       | <b>MICROMART FINTECH</b> | 17,017    | 16,805  | Micro Eazy lives here |
| 3003                                       | Micromart Check off      | 2,416     | 1,616   | —                     |
| 3004                                       | Micromart IPF            | 0         | 0       | empty                 |

## BOTH MICRO EAZY PRODUCTS EXIST, ACTIVE, AND ONE OF THEM IS THE BUSY ONE

The blueprint lists *Micro Eazy (base)* as **OPEN — spec not supplied**, and open item #1 for the founder. That question is closed: the product is configured, active, and has been originating loans since 3 August.

| PRODUCTS · ENTITY 3005 · BOTH ISACTIVE = 1 · BOTH ON WORKFLOW 1022 |                                              |               |                     |                         |           |       |
|--------------------------------------------------------------------|----------------------------------------------|---------------|---------------------|-------------------------|-----------|-------|
| ID                                                                 | PRODUCT                                      | PRINCIPAL     | INTEREST            | TENOR                   | MIN SCORE | LOANS |
| 30219                                                              | <b>Micro Eazy</b> <small>ME</small>          | 5,000–100,000 | 8.25% flat / week   | customer picks, ≤10 wks | 500       | 35    |
| 30220                                                              | <b>Micro Eazy Monthly</b> <small>MEM</small> | 5,000–100,000 | 22.00% flat / month | 2 months                | 500       | 2     |

The base product's tenor is the interesting part. `RepaymentPeriod = 10` is a *ceiling*, not a term: `sp_InsertLoan` takes a `@SelectedPeriod` and the live book uses 4–8 weeks, so interest lands between 33% and 66% rather than a fixed 82.5%. Micro Eazy base is a flexible-tenor weekly product. That is a materially different thing from what the blueprint assumed.

## THE WORKFLOW, AND WHO APPROVES ON IT

Open item #2 — "the Micro Eazy workflow stages, names, order, and who approves at each" — is also answered:

|               |                  |               |                                                            |
|---------------|------------------|---------------|------------------------------------------------------------|
| workflow 1022 | "Micro Eazy"     | EntityID 3005 | created 14 Jul 2026 by user 9093                           |
| stage 2058    | Risk             | root          | CanFinalize 0    AccessID 1    role 41    users 9093, 9098 |
|               |                  |               |                                                            |
|               | v                |               |                                                            |
| stage 2059    | Customer Service | child         | CanFinalize 1    AccessID 2    role 41    users 9093, 9098 |

Two stages, Risk initiates, Customer Service finalises — exactly the shape `scripts/seed-micro-eazy.ts` already builds on our side. That script's stage tree is correct.

THE LOAN BOOK IS REAL MONEY, MOVING YESTERDAY

| LOANS ON PRODUCTS 30219 / 30220 · PROBED 12 AUG 2026 |       |           |             |           |               |
|------------------------------------------------------|-------|-----------|-------------|-----------|---------------|
| PRODUCT                                              | LOANS | PRINCIPAL | OUTSTANDING | DISBURSED | LAST ACTIVITY |
| Micro Eazy                                           | 35    | 306,000   | 386,590     | 32        | 11 Aug, 16:56 |
| Micro Eazy Monthly                                   | 2     | 200,000   | 144,000     | 1         | 5 Aug, 17:46  |

`Transactions.dbo.Disbursements` confirms the money side independently: **33 disbursements, KES 382,000, every one flagged `isSent = 1`**, the most recent yesterday at 16:56:37. The blueprint's "2 loans and KES 144,000 outstanding" is exactly right for Monthly — it simply predates the base product going live and carrying 35 more.

03 Yes — the customers were ported

Confirmed, with the receipt.

`BorrowerEntityMigrationBackup_20260802` holds **17,016 rows**, every one `OldEntityId 3002 → NewEntityId 3005`, every one stamped `MigratedDate 2026-08-02 22:43:07`. A single clean sweep, backed up before it ran. Entity 3005's present count is 17,017 — the 17,016 migrated, plus one account created since. That one is yours.

Their history came with them: entity 3005 now carries roughly 61,000 historical loans against products that still belong to entity 3002 ( `4 WEEKS`, `6 WEEKS`, `JENGA BIASHARA` and so on). Harmless for reporting, but worth knowing before anyone writes a query that joins loans to products on `EntityId` and expects them to match.

04 Your account: found, active, and blocked three ways

0758517032 · EMMANUEL BIRGEN · created 11 Aug 2026, 23:30:28

It is there, in the right entity, and it is a real active record:

|                                                                     |                 |                                               |                         |
|---------------------------------------------------------------------|-----------------|-----------------------------------------------|-------------------------|
| Borrowers.ID                                                        | 168346          |                                               |                         |
| Name                                                                | Emmanuel Birgen |                                               |                         |
| AccountNo                                                           | 254758517032    | NationalID                                    | 39362808                |
| Phone                                                               | 254758517032    | Email                                         | kipletinge123@gmail.com |
| DOB                                                                 | 2002-05-05      | CreditScore                                   | 2000.00                 |
| EntityId                                                            | 3005            | MICROMART FINTECH                             |                         |
| EntityUnit                                                          | 1               | <-- Head Office, which belongs to entity 3002 |                         |
| EntityAgent                                                         | 9096            | AccountStatus 1 (active)                      |                         |
| LoanLimit                                                           | 10,000.00       |                                               |                         |
| Approved / IsROApproved / IsRegistrationCompleted / KycVerification |                 | ALL NULL                                      |                         |
| UssdPin / MobileAppPassword / GeoLocation                           |                 | ALL EMPTY                                     |                         |
| Loans                                                               |                 | NONE                                          |                         |

I then called Micromart's own gate function directly against your borrower ID, for the amounts that matter. It declines all three:

```
dbo.LoanValidation(168346, 30220, 25000) -> 3 "Requested amount exceeds loan limit"
dbo.LoanValidation(168346, 30220, 10000) -> 2 "Insufficient account balance to cover loan charges"
dbo.LoanValidation(168346, 30219, 5000) -> 2 "Insufficient account balance to cover loan charges"
```

`sp_InsertLoan` calls that function first and writes a `DeclinedLoans` row with HTTP-ish code 400 if it returns anything non-zero. So as of right now, **your account cannot be booked a Micro Eazy loan at any amount**. Three distinct causes:

### **BLOCKER** Your record sits on an org unit that cannot see the products

Both Micro Eazy products are scoped `EntityUnits = "0,129"`. Unit 129 is "Main Office", the org unit belonging to entity 3005. Your record is on **unit 1** — "Head Office", which belongs to entity 3002. You are the *only* borrower in the entire Fintech entity on unit 1.

```
EntityUnit distribution, Borrowers WHERE EntityId = 3005
unit 129  17,016 borrowers  <- every migrated customer
unit   1      1 borrower   <- you
```

```
EntityUnit of the 34 borrowers holding Micro Eazy loans
unit 129  34    (100%)
```

**Fix:** move borrower 168346 to `EntityUnit 129`. One field.

### BLOCKER No wallet balance to draw the charges from

`LoanValidation` reads `Transactions.dbo.AccountSavings` and refuses the loan unless the balance covers the pre-paid charges. You have no row there at all. Every real Micro Eazy borrower does — and the debits line up with their bookings.

```
borrower 9738 Amount 150.00 LastAmountTransacted 950.00 11 Aug 16:34 (loan disbursed 16:56)
borrower 152616 Amount 150.00 LastAmountTransacted 950.00 11 Aug 15:24
borrower 148278 Amount 152.50 LastAmountTransacted 400.00 11 Aug 13:29
borrower 168346 -- no row --
```

**Fix:** fund the wallet before the demo. For a KES 25,000 Monthly loan the charges are 1,500 + 100 + 50 = **KES 1,650**.

### BLOCKER Loan limit is 10,000 — below the demo amount

`LoanLimit = 10,000.00`, and the gate rejects any principal above it outright. The blueprint's demo is built on KES 25,000. Raise the limit, or change the demo number — but they have to agree.

### MONEY SAFETY Your phone number resolves to a different person in entity 3002

Borrower `89296` — "Mr Kipleting", NationalID 301968 — carries the same phone *and* the same account number in entity 3002, and is active. Our bridge finds borrowers by phone filtered on entity. Point it at 3002 and it hands back Kipleting, and the loan books against the wrong human being.

```
Borrowers WHERE PhoneNumber = '254758517032'
168346 Emmanuel Birgen NID 39362808 EntityId 3005 active
89296 Mr Kipleting NID 301968 EntityId 3002 active

phones present in BOTH 3002 and 3005: 13
```

Thirteen phone numbers are duplicated across the two entities, so this is a class of bug, not a one-off. Also unfinished on your record: registration was never completed, no KYC, no PIN, no location pin — and the location pin is a hard disbursement gate.

## 05 The misalignment that matters most

The LMS posts to a "Micromart Fintech" that has nothing to do with this one.

`src/lib/enterprise/connections.ts` declares a `micromart-fintech` org and routes all Micromart postings to it via `POSTING_TARGETS`. Its documented target is Techcrast's box at `102.214.69.233,4410`, `EntityId 7`, product `31418` "MIROMART FINTECH", workflow `55` "FINTECH APPROVAL".

I checked every one of those against the live server:

```
Products.ID      31418 -> does not exist here
ApprovalWorkflow 55    -> does not exist here
BsEntity         7     -> no row (7 orphaned borrower records, no entity)
```

So there are two different "Micromart Fintech" deployments, and the LMS is aimed at the one where nothing is happening. `scripts/rehearse-fintech-booking.ts` inherits this: it books against product name `"MIROMART FINTECH"`, capped at KES 5,000–20,000, and describes landing in workflow 55. Every rehearsal we have run proved a pipe into the wrong building.

#### THE GOOD NEWS, AND IT IS GENUINELY GOOD

`sp_InsertLoan` line 54 does this: `SET @Entity = (SELECT EntityId FROM Products WHERE ID = @ProductId)`. It *overwrites* whatever entity the caller passes, and it takes the workflow from `Products.WorkflowId`.

Which means posting product **30220** lands in entity 3005, on workflow 1022, at root stage 2058 "Risk", with `isApproved = 0` — automatically. The demo moment the blueprint describes in step 6 works with *no* change to the posting logic. We only have to get the product ID and the borrower lookup right.

### THREE CONCRETE DEFECTS IN THE BRIDGE AS WRITTEN

#### WRONG PERSON Borrower lookup is entity-scoped, and the entity is wrong

`findBorrowerByPhone` and `ensureBorrower` filter `b.EntityId = @entityId` from `getIdEntityId(org)`, which defaults to **7** for `micromart-fintech`. Two bad outcomes, depending on the deployed env value: at 3002 it returns Kipleting; at 7 it finds nobody, calls `sp_NewBorrowerRegistration` with `EntityId 7`, and books an entity-3005 loan against an entity-7 borrower. Set `MICROMART_FINTECH_ENTITYID = 3005`, and add a national-ID confirmation before posting.

#### WRONG PRICE @SelectedPeriod is never passed

`postLoan` omits it, so the base product falls back to its 10-week ceiling — 82.5% interest on a book that actually prices at 33–66%. Every bridge-posted Micro Eazy loan would be the most expensive one available. The parameter exists; we just have to send the customer's chosen tenor.

#### ATTRIBUTION Channel tag says 7, the live book says 1

`.env.example` sets `LMS_SERVICESUITE_CHANNEL="7"`. All 37 existing Micro Eazy loans carry `ChannelUsed = 1` — which independently confirms **not one loan has ever arrived through our bridge**. Keep 7: it is how we will prove attribution when they do.

## 06 The integration layer, honestly assessed

You asked how Micromart's SMS, STK and email actually work. Two of the five rails work. Two are dead. One is a gate nobody documented.

## STK collections — and only for entity 3005

`Models/Payments.cs` calls `sp_GetUssdStkParams`, which reads `Transactions.dbo.Stkparams` — *not* the empty `StkParams` table in Serviceconnect. That table holds exactly one row, and it belongs to entity 3005: consumer key, secret, passkey and shortcode all present (AES-encrypted, decrypted in `Decipher.cs`), against `https://api.safaricom.co.ke` — live Daraja, not sandbox.

```
Transactions.dbo.StkResponse - real Safaricom receipts
UH57X1SC07  KES 1.00  254706491124  5 Aug 2026 12:25  ResultCode 0
UH5GR1KT7E  KES 5.00  254795595914  5 Aug 2026 13:07  ResultCode 0
```

**The Fintech entity is the only M-Pesa-enabled entity on this server.** Entity 3002 has no `Stkparams` row at all, which is why all 130 of its queued STK requests are unprocessed. Paybill for 3005 is `4116125`.

## WORKS B2C disbursement

33 disbursements totalling KES 382,000 against the Micro Eazy products, all `isSent = 1`, latest 11 Aug 16:56:37. `Transactions` carries 322,070 disbursement records and 225,422 B2C responses estate-wide. This rail is mature and we should not touch it.

## DEAD SMS — nothing sent in over three years

The `SMS` table has 19,438 rows. Every one has `EntityId = NULL`, `companyid = 1`, and the newest was created **3 May 2023**. There are no templates for entity 3005. The Africa's Talking credential row is scoped to the *company*, not the entity, and is flagged `isRegistered = 0`. `SmsProvider` is empty.

Decisively: `Models/SmsManager.cs` contains only read methods — templates, placeholders, campaign lists — and the string `africastalking` appears **nowhere in the entire C# codebase**. The portal manages SMS; it has never sent one. Whatever did the sending is gone.

```
SMS GROUP BY EntityId, companyid, smsprovider
EntityId NULL  companyid 1  provider 1  19,438 rows  last created 2023-05-03 13:46:58
SmsTemplate    10 rows - 7 on entity 3002, 1 on entity 2, 2 unscoped, none on 3005
africaIsTalkingCredentials  1 row, companyid 1, sender "rhs", isRegistered = 0
```

## DEAD Email — never actually dispatched

`Mails` has 46 rows, last created 15 Aug 2023, and `DateSend` is **NULL on every single one**. The table is not referenced anywhere in the C# code. `System.Net.Mail` is imported in three files but no SMTP transport is configured.



## The pre-funded wallet — Micro Eazy's real front door

This is the mechanic nobody wrote down, and it changes the customer journey. A Micro Eazy loan cannot be *created* until the borrower's `AccountSavings` balance covers the pre-paid charges. The customer pays before they borrow. 57,171 borrowers hold wallet rows; the debits happen at booking.

The blueprint's demo script has no top-up step between consent and offer. On stage, that is a code-400 decline in front of the board. This has to be designed into the app flow — and it is arguably a *feature* to show them: charges collected up front, by STK, on a rail that already works.

### WHICH SIDE SHOULD OWN WHAT

Here is the part that makes the "fully operational" goal tractable. The LMS already has, built and per-tenant, precisely the two rails ServiceSuite lost:

| CAPABILITY OWNERSHIP · THE PROPOSED DIVISION OF LABOUR |                              |                                         |                |
|--------------------------------------------------------|------------------------------|-----------------------------------------|----------------|
| RAIL                                                   | SERVICESUITE (3005)          | BIRGENAI LMS                            | OWNER          |
| STK collections                                        | LIVE PROD                    | mpesa/daraja.ts per-org vault           | ServiceSuite   |
| B2C disbursement                                       | 33 SENT                      | initiateB2C + maker-checker             | ServiceSuite   |
| SMS                                                    | DEAD 2023                    | sms/send.ts — AT, queued, billed, EN/SW | LMS            |
| Email                                                  | NEVER SENT                   | email/send.ts — nodemailer per-org      | LMS            |
| Loan booking                                           | sp_InsertLoan → wf 1022      | postLoan bridge                         | ServiceSuite   |
| Scoring · CRB · decisioning                            | min score field only         | 7-stage engine, Metropol, v3.1.1        | LMS            |
| Credentials                                            | encrypted, per entity, in DB | OrgIntegration vault, per org           | both, mirrored |

The LMS vault already defines the exact integration kinds needed — `MPESA_STK`, `MPESA_B2C`, `MPESA_C2B`, `SMS`, `CRB`, `SERVICESUITE` — encrypted per organisation with status tracking. So "fully operational" is mostly a wiring job into a socket that exists, not new construction.

### 07 Five corrections the board document needs

The blueprint was accurate on 5 August. The system moved. These are the deltas that would embarrass us in the room.

#### CORRECTION 1 The worked example on page one is wrong

§8.2 says a KES 25,000 loan nets the customer 23,500 after the 6% fee is "deducted before disbursement". It is not. The fee's `FeeType = 1` ("Before Disbursement") means *collected separately*; only `FeeType = 2` reduces the payout. `sp_InsertLoan` computes `AmountToDisburse = @Principal - @DeductedCharges`, and deducted charges are zero.

```
Live proof - loan 441233, Micro Eazy Monthly
Principal 100,000.00   Interest 44,000.00   LoanAmount 144,000.00
AmountToDisburse 100,000.00   <- the full principal, not net of the 6,000 fee

Corrected worked example, KES 25,000:
Disbursed to customer   25,000   (full principal)
Charges, paid up front   1,650   PF 1,500 + CRB 100 + Security 50
Total repayable         36,000   2 x 18,000   <- blueprint is right here
```

#### CORRECTION 2 There are three mandatory charges, not one

The blueprint documents the 6% processing fee only. Both products also carry a **CRB Fee of KES 100** and a **Security Fee of KES 50**, both fixed, both `IsActive = 1` and `IsMandatory = 1`, created 3–5 August. Our offer card must show all three or the arithmetic will not match theirs.

#### CORRECTION 3 The 650–6,000 fee cap is decorative

The product stores `MinValue 650 / MaxValue 6,000`, and the blueprint quotes them. Neither `sp_InsertLoan` nor `LoanValidation` reads those columns — both compute a flat percentage. A KES 5,000 loan is charged **300**, not the 650 floor. Do not quote the cap to the board until it is enforced, or the first small loan contradicts us.

#### CORRECTION 4 "Workflow 1021" is the wrong workflow

§4 and the architecture diagram show the bridge posting to `wf 1021`. Workflow 1021 is "APPROVAL WORKFLOW" and it belongs to **entity 3002** (Officer Review → Final Approval). Micro Eazy is `wf 1022` in entity 3005. The diagram should say 1022 — and note that the product, not the caller, selects it.

#### CORRECTION 5 Two open items are closed, and the typo is not ours

Founder open items #1 (base product spec) and #2 (workflow stages) are both answered above — delete them. Sprint 0 tasks 0.2 and 0.3 are done on the ServiceSuite side; what remains is mirroring them into our Postgres. And Sprint 0 task 0.11, "fix the `MIROMART FINTECH` typo": that string does not exist on this server. It lives on the old Techcrast deployment and in our own `connections.ts` comments. The live entity is spelled `MICROMART FINTECH` correctly — though it is in caps, and its contact fields are placeholder junk ( `EntityTaxRef "kdksk"`, `EntityAddress "hdhdh"`, email `infp@` for `info@` ).

## 08 Where the Super LMS actually stands

Realm A is as strong as claimed. The spine and the app are untouched.

I verified Sprint 0 against the working tree rather than the document. Note the tree has no `.env` or `.env.local` — only `.env.example` — so I cannot read deployed values, and nothing below asserts what production is running.

| SPRINT 0 · DEMO-CRITICAL · VERIFIED IN THE REPO 12 AUG 2026 |                        |              |                                                                                                                                                                        |
|-------------------------------------------------------------|------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| #                                                           | TASK                   | STATE        | EVIDENCE                                                                                                                                                               |
| 0.1                                                         | Micro Eazy logo        | PART         | 6 SVGs in <code>public/brand/micro-eazy/</code> ; no PNG raster set                                                                                                    |
| 0.2                                                         | Seed both products     | READY        | <code>seed-micro-eazy.ts</code> exists; needs <code>--ss-me=30219 --ss-mem=30220</code>                                                                                |
| 0.3                                                         | Mirror the workflow    | CORRECT      | Risk → Customer Service matches wf 1022 exactly                                                                                                                        |
| 0.4                                                         | Exchange v1            | NOT STARTED  | no <code>MarketplaceListing</code> / <code>AllocationPolicy</code> / <code>AllocationDecision</code> in <code>schema.prisma</code> ; no <code>src/lib/exchange/</code> |
| 0.5                                                         | PWA shell              | NOT STARTED  | no <code>src/app/manifest.ts</code> , no <code>public/sw.js</code>                                                                                                     |
| 0.6                                                         | microeazy host routing | NOT STARTED  | no <code>microeazy</code> reference in middleware or branding                                                                                                          |
| 0.7                                                         | Hub tile               | OTHER REPO   | not in this tree                                                                                                                                                       |
| 0.8                                                         | Why-declined screen    | NOT STARTED  | engine emits reason codes; no customer surface                                                                                                                         |
| 0.9                                                         | Limit ladder           | NOT STARTED  | <code>GraduationEvent</code> exists server-side only                                                                                                                   |
| 0.10                                                        | Bridge rehearsal       | WRONG TARGET | <code>rehearse-fintech-booking.ts</code> aims at product 31418 / wf 55                                                                                                 |
| 0.11                                                        | Fix the typo           | N/A          | string absent from this server — see correction 5                                                                                                                      |

`verify-micro-eazy.ts` is the encouraging one: it already hardcodes the expected `ssId 30219` and `30220` with rates 82.5 and 44. Somebody had already found the live product IDs. What was missing is that the connection those IDs live behind is not the one the posting target points at.

## 09 How we make it possible — for every lender

Sequenced by dependency. Written per-platform, not per-lender, because every defect below is structural.

### THE CHARGE COLLECTION, EXACTLY AS IT MUST RUN

Decision 4 approved: our app collects the charges by STK before the offer is signed. I traced the wallet procedures to get the sequence right, and there is a trap in them.

## THE TRAP — THIS ONE SILENTLY EATS THE CUSTOMER'S MONEY

`sp_UpdateSavings` is a bare `UPDATE`. For a borrower with no `AccountSavings` row — which is *every* customer our pipeline originates, including yours — it matches zero rows, succeeds, and credits nothing. The STK collection lands in the lender's paybill, the wallet stays at zero, and the loan still declines with code 2.

**Use `sp_CreateAccountSavings` instead.** It inserts when the row is missing and credits when it exists, and it writes the matching `CustomerStatement` line so the top-up is visible to the customer and the officer.

```
1. customer accepts terms
2. compute charges = SUM(fixed FeeType=1 fees) + (SUM(pct FeeType=1 fees)/100 x Principal)
   ^ use the LoanValidation formula, NOT sp_InsertLoan's - see below
3. STK push for that amount, on THAT lender's own Daraja credentials
4. wait for callback ResultCode = 0 + M-Pesa receipt
5. EXEC sp_CreateAccountSavings @BorrowerId, @amount, 'Micro Eazy charges - '
6. re-assert dbo.LoanValidation(borrower, product, principal) = 0 <-- hard gate, never skip
7. sp_InsertLoan -> lands at the lender's root stage, isApproved = 0
8. their officer approves in their own system
9. sp_InsertUpfrontLoanCharges debits the wallet at approval -> we must NOT debit it ourselves
```

Step 2 matters more than it looks. Two different charge formulas exist in the database: `LoanValidation` sums *all* pre-disbursement fees and always multiplies percentages by principal, while `sp_InsertLoan` counts only mandatory fees and varies the base by `FeeRef`. **The gate is `LoanValidation`, so top up to its number** — if we use the other one on a lender whose product has optional fees or an interest-based fee, we under-collect and the loan declines at booking.

I checked all 23 active products on this server and the two formulas currently agree, so Micromart will not expose the difference. Axe and Techcrast are on other servers with their own fee tables, and they might.

## BLOCK A — MAKE THE REGISTRY TRUSTWORTHY, THEN PROVE IT

1. **Delete `POSTING_TARGETS` and `getPostingOrg()`**. Decision 3 retires the only deployment that needed a cross-server split, so posting goes where reading goes. This removes the exact mechanism that pointed us at a dead box.
2. **Move the bridge config into the vault** as `kind: SERVICESUITE`. `ServiceSuiteConfig` already exists in `src/lib/vault/integrations.ts` with `connectionString`, `entityId`, `postingEnabled`, `createdByUserId` and `channel` — it is simply never read. Add `entityUnitId`, then have `getMssqlConfig()` resolve from the vault with the env vars as fallback. Connection and entity ID travel together from that point on, which is what kills the `axe = 3003` collision.
3. Seed our side with the real IDs: `seed-micro-eazy.ts --ss-me=30219 --ss-mem=30220`, then run `verify-micro-eazy.ts` and expect it green.
4. **Fund the wallet once, by hand, for the rehearsal** — KES 1,650 via `sp_CreateAccountSavings` — then require `LoanValidation = 0` before anyone posts. The automated path from the sequence above replaces this for real customers.
5. Retarget the rehearsal script at product 30220 and workflow 1022, generalise it to take a lender slug, and rehearse twice with a reversible test borrower — the blueprint's own rule, which we have never actually satisfied against this server.

## BLOCK B — CLOSE THE SAFETY GAPS THE BRIDGE HAS TODAY

1. Add a national-ID confirmation to `findBorrowerByPhone` before any post. Thirteen duplicated phones across 3002/3005 is thirteen chances to lend to the wrong person, and yours is one of them.
2. Plumb `@SelectedPeriod` through `postLoan` so the customer's chosen tenor prices the loan, instead of silently defaulting to the most expensive term.
3. Surface all three charges and the wallet pre-fund step in the offer card, and add the top-up to the app flow — collected by STK on the rail that already works.
4. Move the bridge config into the `OrgIntegration` vault as `kind: SERVICESUITE`. It is env-driven today, which is exactly how we ended up pointed at a dead deployment without noticing.

## BLOCK C — TAKE OWNERSHIP OF WHAT SERVICESUITE LOST

1. Wire `src/lib/sms/send.ts` to the Micromart org and let the LMS own every customer message: approved, disbursed, payment received, T-3/T-1/T0. ServiceSuite has not sent an SMS since May 2023, so there is nothing to conflict with and nothing to migrate.
2. Same for email via `src/lib/email/send.ts`. The `Mails` table has never dispatched anything.
3. Register the Africa's Talking sender properly — the existing credential row is `isRegistered = 0` under sender ID "rhs", which is not a name a Micro Eazy customer should ever see.
4. Keep the standing rule from the blueprint's risk table: blank the SMS credentials in every verification and rehearsal script. Now that we are about to make SMS actually work, that rule stops being theoretical.

## BLOCK D — THE BUILD THE BLUEPRINT IS ACTUALLY ABOUT

The Exchange (0.4) and the PWA (0.5, 0.6, 0.8, 0.9) are untouched and unblocked — none of them depend on anything above. But they are new construction, and Blocks A and B are the ones standing between us and a loan appearing in Micromart's own Micro Eazy queue. Do those first; they are days of configuration, not weeks of engineering.

### THE REFRAME WORTH CARRYING INTO THE ROOM

The blueprint's pitch was "you built Micro Eazy, we built the machine that fills it." That has gotten stronger, not weaker. They have gone from 2 loans on one product to **37 loans and KES 382,000 disbursed across two**, on a production M-Pesa rail, with 17,016 customers moved into the entity on purpose. They are not evaluating whether to try this. They are already running it, by hand, at small scale, with no pipeline and no notifications.

That is a better door than the one the document described.

---

## 10 The prerequisite: a working machine

None of section 09 can be written on this laptop yet. The repo was pulled fresh, so there is no `node_modules`, no Node runtime, and no `.env.local` — the secrets were never committed (`.gitignore` line 34 correctly ignores `.env*`, and git history confirms they never landed). That is the right hygiene, and it is also why the toolchain has to be rebuilt before any code lands.

**Node does not need a virtual environment.** `node_modules` is already per-project isolation — the Python `venv` concept has no equivalent here and adding one would be overhead with no benefit. What is missing is just the runtime and the install.

### THE ONE SECRET THAT IS NOT REPLACEABLE

#### CHECK THIS BEFORE ANYTHING ELSE

`VAULT_MASTER_KEY` is the AES-256-GCM key for `OrgIntegration.configEnc` — every lender's ServiceSuite connection, Daraja credentials, SMS key and CRB key is encrypted in Postgres with it. The database is on Supabase, so the *data* survived the laptop change. The key did not, unless it is in a password manager or the Vercel environment.

If that key is gone, every vault row is unrecoverable ciphertext and each lender's credentials must be re-entered by hand. Everything else in `.env.example` is either recoverable or degrades gracefully by design.

**Confirm you still hold it first.**

### MINIMUM TO BOOT AND WORK ON THE BRIDGE

Most of `.env.example` is optional by construction — storage falls back to simulation, each KYC leg is independent, billing no-ops, and the route map says so on screen. Five variables get the app running:

```
DATABASE_URL      Supabase pooler           } the app will not boot without these
DIRECT_URL        Supabase session pooler }
NEXTAUTH_SECRET   any 32-byte random
NEXTAUTH_URL      http://localhost:3000
VAULT_MASTER_KEY   32 bytes hex                 <- must be the ORIGINAL, see above

then, for the Micromart bridge work specifically:
LMS_POSTING_ENABLED "false"             <- KEEP FALSE LOCALLY. posting is armed at production.
LMS_SERVICESUITE_CREATED_BY UserMaster.ID of the BirgenAI service account
SERVICESUITE_CONN_MICROMART the Tailscale string (until Block A moves it to the vault)
```

One limitation to plan around: Daraja rejects any callback URL that is not public HTTPS, so **the STK top-up from decision 4 cannot be tested end-to-end from localhost**. It needs `PUBLIC_BASE_URL` pointed at a deployed preview or a tunnel. The charge maths and the `sp_CreateAccountSavings` call can both be verified locally; only the phone-rings moment needs the public address.

## THE SEQUENCE

- 1 Install Node 22 LTS — `winget install OpenJS.NodeJS.LTS` (winget is available on this machine; nothing else is needed).
- 2 `npm ci` — not `npm install`. There is a `package-lock.json`, and `ci` reproduces it exactly instead of silently resolving new versions of Next 16 / Prisma 7 / React 19.
- 3 Recreate `.env.local` from the block above.
- 4 `npx prisma generate`, then `npm run dev`.
- 5 `npx tsx scripts/verify-micro-eazy.ts` — the fastest honest check that the LMS and the live book agree.

Optional: `nvm-windows` if the Hub and the agent projects need different Node versions than this one. Skip it if they do not — a single Node 22 is simpler.

Every figure in this report came from the live database, probed 12 Aug 2026 — no writes were made from here. Blueprint claims were re-probed rather than repeated; where this document and `reports/Micro-Eazy-Ecosystem-Blueprint.docx` disagree, section 07 explains why. No application code has been changed: with no Node runtime on this machine nothing could be typechecked, and unverified changes to a bridge that posts into a production loan book are not worth handing over.